

Homework 7: Relational Algebra and Triggers

Assigned on: Mar 5

Due on: Mar 13

Exercise 1: Relational Algebra (50 points)

Consider the relational schema of a Hospital Management System shown in Figure 1. Write a relational algebra query for each of the following:

1. Get a list of average doctors' salary for each department
2. Get a list of patients' names who are operated on by "Dr. Smith" on the date "10-23-2022"
3. Get a list of all doctors' names who do not have any patients

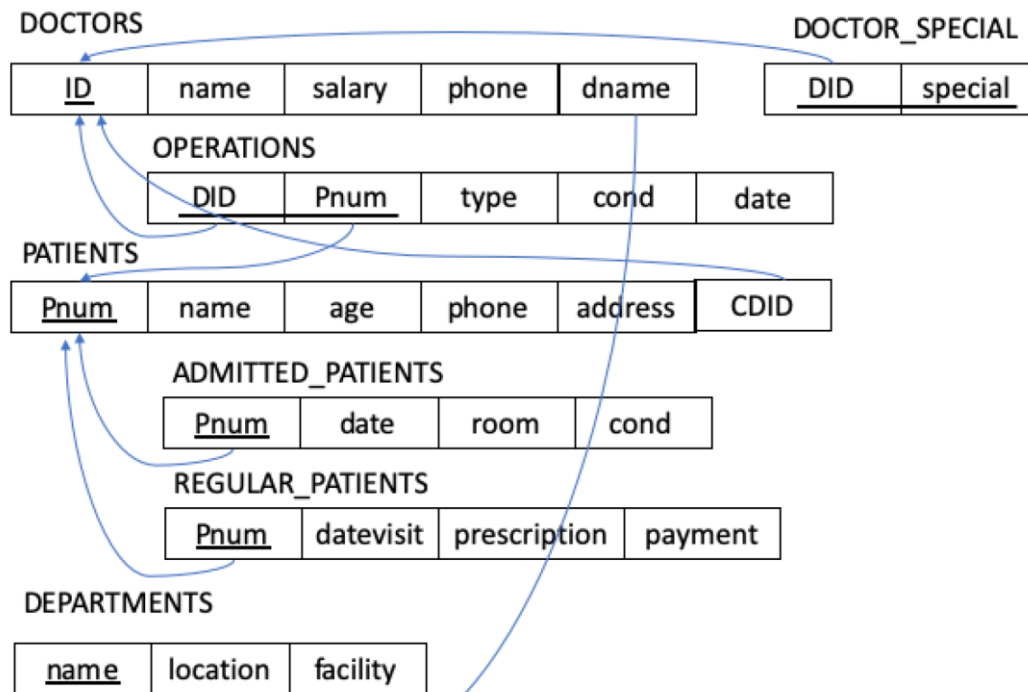


Figure 1: Schema for Exercisse 1

Exercise 2: Trigger (50 points)

Create the following tables

- **student_mast** which includes 3 columns **STUDENT_ID**, **NAME**, **ST_CLASS**.
- **stu_log** which includes 2 columns **user_id** and **description**.

Create a Trigger for **AFTER UPDATE** which will add to the **stu_log** table an entry for each change in a row in table **student_mast**. For example, if the following **UPDATE** was executed

To the table shown in Table 1, the result will be as shown in Table 2 and Table 3.

UPDATE student_mast SET ST_CLASS = ST_CLASS + 1;

Table 1: Initial State of student_mast table

STUDENT_ID	NAME	ST_CLASS
1	Alice	7
2	Bob	8

Table2: State of student_mast table after update

STUDENT_ID	NAME	ST_CLASS
1	Alice	8
2	Bob	9

Table3: State of stu_log table after update

USER_ID	DESCRIPTION
1	1 student promoted from class 7 to 8
2	2 student promoted from class 8 to 9